

An analysis of patterns of human-caused versus lightning-caused fires

By Somesh Joshi

Introduction:

Wildfires are an ever-increasing concern in forest management, especially in regions like British Columbia, where the frequency and intensity of these fires have been increasing due to climate change and other human activities. In 2018, we saw a record-breaking 2117 fires, the damages from which exceeded 600 million dollars (“Wildfire Averages”). The Okanagan basin, known for its diverse ecosystems and local townships, has experienced some of the largest fire seasons on record, most notably in 2017 and 2018. Understanding the spatial patterns of forest fire occurrence, annual area burned, and fire size is essential for effective resource management and long-term planning. This research aims to compare the spatial patterns of human-caused versus lightning-caused fires in the Okanagan basin during a single year (2018). Geographic Information Systems (GIS) plays a crucial role in this analysis, offering tools for mapping, analyzing, and visualizing wildfire data to uncover patterns and trends. By examining the spatial-temporal patterns of wildfires, this study seeks to provide insights into the different drivers of fire occurrences and their impacts on the landscape. The importance of this research lies in its potential to inform targeted fire management strategies, enhance preparedness, and mitigate the adverse effects of wildfires on ecosystems and communities.

Study Area:

The Okanagan is a region in the southern part of British Columbia known for its dry, warm climate and diverse topography, including forests, valleys, and mountains. This area, while renowned for its wine and fruit production, is also vulnerable to wildfires due to factors such as extended periods of dry weather, frequent lightning storms, and human activities (“Expert Q&A”). Wildfires can be sparked by lightning strikes in remote, forested areas or by human actions such as unattended campfires, discarded cigarettes, or deliberate acts of arson, particularly near roads and populated areas where human presence is high (“What Causes Wildfires”). Below is a map of our study area (Figure 1).

Study Map

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Projection: GCS_North_American_1983

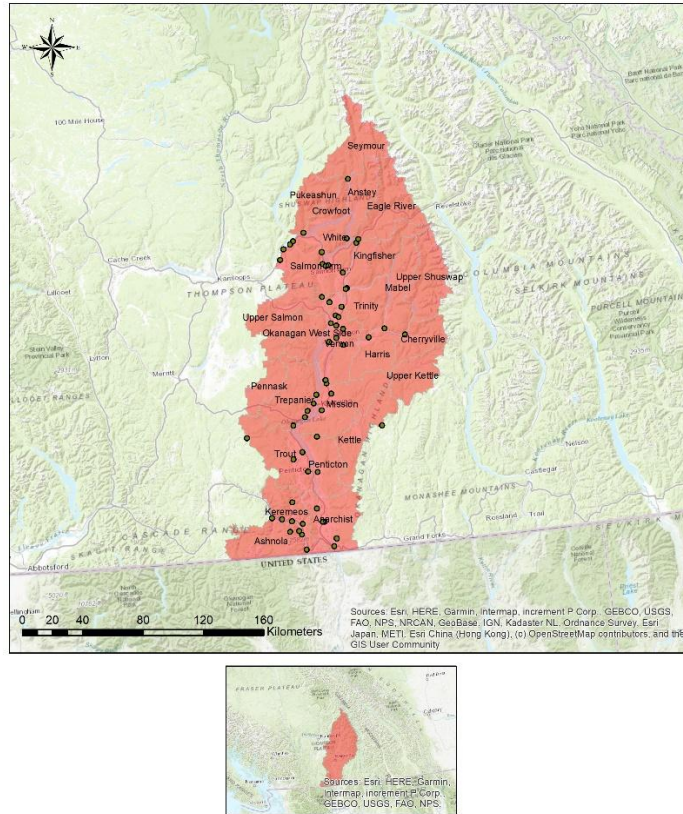


Figure 1

Spatial Distribution:

The Mean Center tool in ArcGIS is a spatial analysis method that computes the geographic center of a set of points. This center is determined by calculating the average x -coordinate and the average y -coordinate of all the points in the dataset, effectively finding the balance point within the dataset where all points have an equal weight.

In our study, this tool is used to identify the central tendency of two different types of fire origins in the Okanagan region for the year 2018. To accomplish this, the original dataset “ok_fire_2008_2018.shp” was divided into two subsets: one representing the locations of

lightning-caused fires and the other representing the locations of person-caused fires. The Mean Center tool was then applied to each subset separately, yielding two mean center points that were recorded in the output file “fire_mc.shp”.

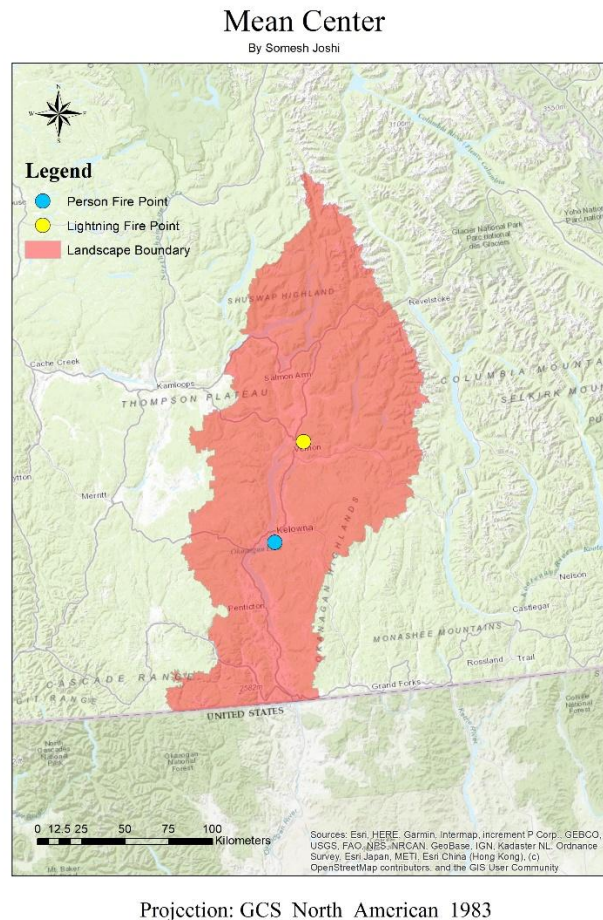


Figure 2

The resulting map (Figure 2) displays two distinct points: one for the mean center of lightning-caused fires (yellow) and one for the mean center of person-caused fires (blue). These points are overlaid on the map of the Okanagan region within its landscape boundary.

Analyzing the map, we observe that the mean center for person-caused fires is near Kelowna, a major urban center in the Okanagan, suggesting that person-caused fires may be associated with areas of higher population density and human activity. In contrast, the mean center for lightning-caused fires is located closer to Salmon Arm, which might indicate that these fires occur more

frequently in areas that are less influenced by human activity, reflecting a more natural fire regime.

The comparison of the two mean centers helps in understanding how different factors contribute to fire risks in the Okanagan. It provides a basis for further investigation into the underlying causes of fires and aids in designing targeted fire management and prevention strategies. For instance, resources for mitigating person-caused fires could be focused around urban and recreational areas, while strategies for lightning-caused fires might be more about early detection and rapid response in remote locations.

Spatial Point Pattern Analysis:

Next, we used the Average Nearest Neighbor tool in ArcGIS which is designed to quantify the spatial pattern of point features within a study area. It evaluates whether the pattern expressed by the points is clustered, dispersed, or random. This is done by calculating the average distance from each point to its nearest neighboring point and comparing it to what would be expected under a random distribution. The degree of clustering or dispersion is indicated by the nearest neighbor ratio, with values less than 1 suggesting clustering, values greater than 1 indicating dispersion, and a value of 1 representing a random distribution.

The tool was applied to two datasets as mentioned previously: one representing lightning-caused fires and the other person-caused fires, both from the year 2018. The results include the nearest neighbor ratio (NNRatio), the z -score, and the p -value. The z -score and p -value help determine the statistical significance of the spatial patterns observed.

For person-caused fires, the NNRatio was 0.69, with a z -score of -4.71 and a highly significant p -value of 0.000003. This indicates a strong clustering pattern of person-caused fires, much different from a random distribution. The observed average nearest neighbor distance was shorter (6460.19 m) than the expected distance under a random pattern (9330.02 m), reinforcing this clustering pattern.

The lightning-caused fires also exhibited a clustered pattern with an NNRatio of 0.85. While this ratio is closer to 1 than that of person-caused fires, it still indicates clustering, as supported by a z -score of -3.43 and a significant p -value of 0.000611. The observed average nearest neighbor

distance for lightning-caused fires was 5918.45 m, shorter than the expected distance of 6956.91 m, though the degree of clustering was not as pronounced as in person-caused fires.

Comparing these findings, we can conclude that both person-caused and lightning-caused fires in 2018 were significantly clustered in the Okanagan region, with person-caused fires showing a stronger clustering pattern. This information could suggest that person-caused fires may be more influenced by factors associated with specific locations, such as proximity to roads, campgrounds, or urban interfaces. On the other hand, lightning-caused fires, while still clustered, might be influenced by natural landscape features—perhaps those that are conducive to lightning strikes, such as ridgelines or higher elevations.

Global Moran's I:

Moran's I is a statistic used to measure spatial autocorrelation, which quantifies the degree to which similar values in a dataset are located near each other spatially. In the context of fire analysis, it can indicate whether areas with high fire sizes are clustered together or dispersed. A Moran's I value close to +1 suggests positive autocorrelation, meaning that similar high or low fire sizes are clustered. Conversely, a value close to -1 indicates negative autocorrelation, meaning that similar sizes are dispersed. A z-score is used alongside the Moran's I value to assess the statistical significance of these patterns, with a significant positive z-score suggesting clustering and a significant negative z-score suggesting dispersion.

For the study, this tool was utilized to assess the spatial autocorrelation of fire sizes for both lightning-caused and person-caused fires in 2018. The data was aggregated by polygon using the sum of fire sizes within each landscape unit. Spatial relationships were conceptualized based on contiguity, with row standardization being applied (due to the use of aggregated data).

The results showed a Moran's I value of 0.06 for lightning-caused fires, with a positive z-score of 3.52 and a significant *p*-value, indicating a clustering pattern of larger fires. This suggests that larger lightning-caused fires in 2018 were not randomly distributed, but rather tended to occur near each other. On the other hand, person-caused fires showed a Moran's I value of -0.02, with a non-significant positive z-score of 0.63 and a high *p*-value, suggesting no significant pattern of clustering or dispersion.

The implication of these findings is that while lightning-caused fires tend to cluster in certain areas and are potentially influenced by natural landscape features conducive to lightning and fire spread, person-caused fires do not exhibit a significant spatial pattern. These results reinforce the importance of focusing fire management efforts on known areas of lightning-caused fire clusters and suggest that person-caused fires might be more randomly distributed, requiring a different approach to prevention and management.

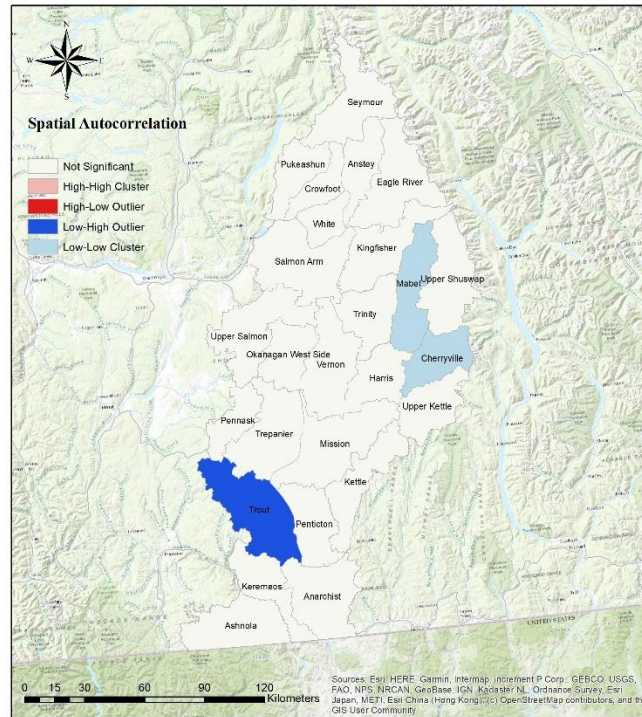
Local Moran's I:

The Cluster and Outlier Analysis tool, leveraging the local Moran's I statistic, scrutinizes the spatial patterns of attributes within a geographic dataset, revealing where values are more similar or dissimilar to their surroundings. This method identifies clusters of similarly high values (high-high) or low values (low-low), as well as spatial outliers where a high value is surrounded by low values (high-low), or a low value by high values (low-high).

Applied to the fire data, we produced two separate maps for person-caused and lightning-caused fires, categorizing areas into significant clusters or outliers based on fire size sums for 2018. For person-caused fires, most polygons showed no significant patterns, but there were clusters where fire sizes were consistently low, and an outlier where a smaller fire was surrounded by areas with larger fires. The results for lightning-caused fires also showed mostly non-significant patterns, but with a notable cluster of larger fires and an area of smaller fires surrounded by larger fire zones.

Person Local Moran's I Map

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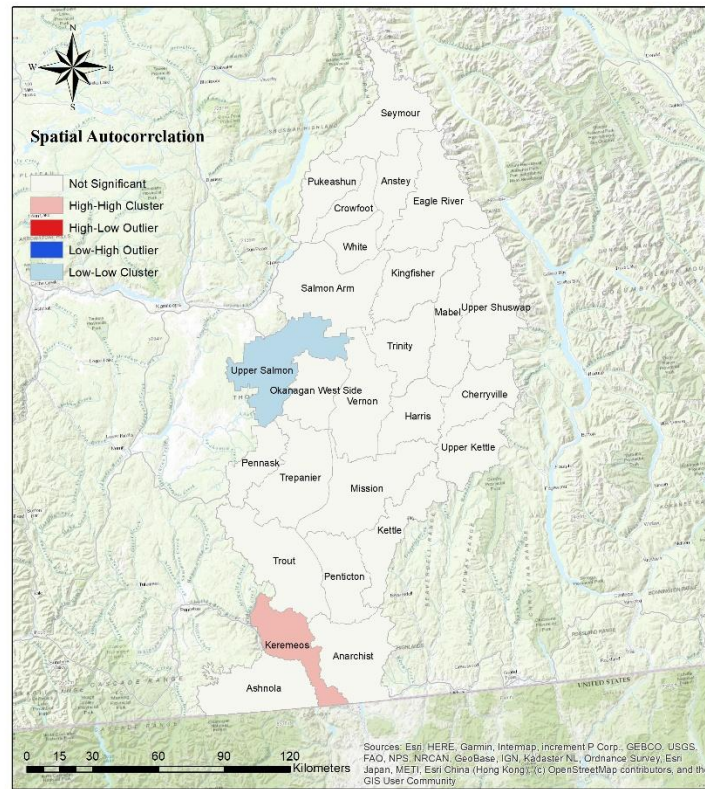
Projection: GCS_North_American_1983

Figure 3

The local Moran's I map for person-caused fires showed significant low-low clusters occurred in Mabel and Cherryville, suggesting that these areas have a pattern of smaller fires which could be related to effective fire prevention and suppression activities or less flammable vegetation. An outlier was found in Trout, indicating an area with small fires surrounded by regions with larger ones, hinting at potentially unique environmental or human factors affecting fire behavior.

Lightning Local Moran's I Map

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Projection: GCS_North_American_1983

Figure 4

For lightning-caused fires, a significant high-high cluster was identified in Keremeos, pointing to a tendency for larger fires in this area, which may be due to prevalent dry conditions, lightning strike frequency, or fuel availability. Conversely, the low-low cluster in Upper Salmon suggests a pattern of smaller fires, potentially due to less severe lightning activity or more effective natural barriers to fire spread.

These spatial insights are crucial for tailoring fire management approaches, suggesting a focus on high-risk areas for resource allocation, and highlighting regions where current strategies might be working effectively. Understanding the specific local dynamics of fire sizes helps in refining predictive models and directing community awareness programs appropriately.

Conclusion:

This research has meticulously highlighted the contrasting spatial patterns and clustering behaviors of human-caused and lightning-caused wildfires in the Okanagan basin, leveraging a suite of GIS tools to decipher the complex dynamics governing these events. Our findings pronounce the clustering of human-caused fires in proximity to urban areas, indicative of the anthropogenic risk factors at play. On the flip side, lightning-caused fires demonstrated clustering in more remote locales, hinting at natural factors influencing fire ignition and spread. The use of spatial statistics, including Moran's I and the Kernel Density Estimation, has unraveled nuanced patterns—from the localized clusters of small, human-induced fires to significant clusters of large, lightning-caused fires. These insights carve out a more detailed understanding of fire ecology in the region and point out the need for diversified fire management strategies. Such strategies should be tailored with person-caused fire prevention focusing on urban education and mitigation, and lightning-caused fire management leaning towards enhancing detection and rapid response in the wildland-urban areas and beyond. As climate change continues to reshape our landscape, this critical insight empowers us to make informed decisions, aiming to safeguard our ecosystems and communities against the looming threat of wildfires.

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